ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

CIT3371 - CRIMINAL INVESTIGATION AND INVESTIGATIVE INTERVIEWING

FINAL EXAMINATION

TIME ALLOWED : **WRITING – THREE HOURS**

: **READING 5 MINUTES**

DATE : **THURSDAY 28 MAY 2026**

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **Any Three (3)** questions from this section.
3. The number of marks assigned to a question should guide the amount of content you provide in your response.
4. This examination paper carries a total of **100 marks**.
5. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A – COMPULSORY

QUESTION ONE (40 Marks)

Case Scenario: "The Ghost in the Pension Fund"

You are a Senior Criminal Investigator in Lusaka, Zambia. A whistle-blower inside the **National Pension Scheme Authority (NAPSA)** has come forward with a serious allegation.

According to the whistle-blower (who you have verified as reliable), a senior IT systems administrator named **Peter Mwansa** has been running a sophisticated fraud scheme for the past 18 months. Peter uses his privileged access to NAPSA's central database to:

- **Create fake "ghost" pension accounts** for non-existent persons.
- **Siphon small amounts (K50 – K100)** from thousands of genuine member accounts into these ghost accounts just before monthly interest is calculated.
- **Transfer the accumulated funds** from ghost accounts into a single mobile money account controlled by his cousin, **Sarah Banda**, a phone vendor in Soweto Market.
- **Delete the daily transaction logs** every 48 hours using an automated script he installed.

The whistle-blower provided you with:

- A printout showing Peter's user ID (P. MWANSA) logged into the NAPSA database at **01:17 AM** on a public holiday.
- A list of 47 suspicious "ghost" accounts all registered with the same inactive mobile number.

You have no physical evidence yet. You also discovered that Peter is scheduled for a two-week leave starting in **6 days**. You suspect he may delete remaining evidence or flee before then.

Your Task:

You are the lead investigator. Produce a **formal Investigations Plan** using the structure below.

Structure your answer as follows:

1. Case Summary, Allegation & Offences (6 marks)

2. Key Parties Involved (4 marks)

3. Investigation Objectives (5 marks)

4. Methodology & Investigative Techniques (15 marks)

Select **three (3)** investigative techniques applicable to this case.

For **each** technique, state:

- Why it is appropriate for this specific case (1 mark)
- How you will execute it, step by step (2 marks)
- What specific evidence you seek (1 mark)
- What legal authority you need (e.g., warrant, court order) (1 mark)

5. The PEACE Model (6 marks)

- You will interview Peter Kwanza after collecting digital evidence. Explain how you will apply the **PEACE model** specifically to him.

6. Risk & Urgent Action (4 marks)

- Identify the **at least on major risk** to this investigation.
- Propose **one specific action** you will take within the next 24 hours to mitigate that risk.

SECTION B: ANSWER ANY THREE (3) QUESTIONS (20 Marks Each)

QUESTION TWO (20 Marks)

Scenario:

A ZCAS university student, **Brian**, reports his new smartphone (worth K4,000) stolen from his library locker. He suspects his classmate, **James**, because James asked him many questions about the phone's password. You are the investigating officer. No CCTV. No witnesses. James denies everything.

Later, Brian uses "Find My Device" and tracks the phone to a house in a nearby compound. A young woman, **Chanda**, answers the door. She admits buying the phone for K800 from "a boy at the bus stop" but does not know his name.

(a) Parties to the Investigation (8 marks)

Identify the **legal role** of each of the following persons in this case using the correct terms

Person	Legal Role	Reason (one sentence)
I. Brian		
II. James (before formal charges)		
III. Chanda		
IV. You		

(b) Types of Interviews (4 marks)

Later, after you find evidence linking James to the theft, you arrest him. You now want to question him again. What **type of interview** must you now use? Name it and explain what you must tell James before starting. **(4 marks)**

(c) Chain of Custody (8 marks)

You seize the phone from Chanda and plan to use it as evidence against James.

Explain what a **chain of custody** is and list **three things you must document** when handling this phone from the moment you take it from Chanda until it is presented in court. Further explain why poor chain of custody may break a good case.

QUESTION THREE (20 Marks)

(a) Compare and **contrast** physical evidence (e.g., fingerprints, DNA, a weapon) with **digital evidence** (e.g., data from a computer, mobile phone, or cloud account). **(8 marks)**

(b) State whether each of the following statements is **TRUE** or **FALSE**. **(2 marks each – Total 12 marks)**

(i) A digital forensic analyst can be qualified as an expert witness in court.

(ii) Locard's Exchange Principle states that "every contact leaves a trace," and this applies to both physical and digital evidence.

(iii) A chain of custody document only needs the signature of the investigator who first collected the evidence.

(iv) In Zambia, a judge has discretion to admit **illegally** obtained evidence.

(v) A forensic image (bit-for-bit copy) of a hard drive is legally considered the same as the original evidence.

(vi) Memory forensics (analysing RAM) is useful because data stored in RAM disappears when a computer is powered off.

QUESTION FOUR (20 Marks)

(a) A convicted cybercriminal is interviewed in prison. He says: "I don't force anyone to click links. I don't hack passwords. People give me their money because they want something for nothing. Greed is my best tool." **Do you agree or disagree with this statement?** Justify your answer using **two (2) different cybercrimes** as examples. **(6 marks)**

(b) In digital forensics, there are four core principles that govern all defensible forensic work. Name and explain each of the four principles **(8 Marks)**

(c) In Zambia we do not have good investigators. Argue for or against this statement. **(6 marks)**

QUESTION FIVE (20 Marks)

(a) Discuss the **importance of surveillance** in criminal investigations. Then, explain the key differences between **physical surveillance** and **electronic. (8 marks)**

(b) You are investigating a series of armed robberies at fuel stations in Lusaka. Your intelligence suggests a specific gang is responsible, but you do not have enough evidence to arrest them. Your supervisor proposes a **sting operation**.

i. What is a **sting operation**? Describe how a sting operation would work for this fuel station robbery case. **(4 marks)**
ii. Identify **two (2) specific risks** of conducting a sting operation in this scenario. For each risk, propose how you would reduce or manage it. **(4 marks)**

(c) You decide to conduct **canvassing** in the neighbourhoods near the robbed fuel stations. Name and briefly explain the **two (2) types of canvassing** discussed in criminal investigations. For each type, state what kind of evidence you hope to obtain. **(4 marks)**

[Total: 20 marks]

END!! ALL THE BEST!!!